

1

Reflex arc components

COMP

You should be able to: Diagram a reflex arc naming the receptor, sensory neuron, integrating center, motor neuron, and effector, and classify the reflex as monosynaptic or polysynaptic.

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2

COMP

Muscle spindle and stretch reflex

You should be able to: Explain how the muscle spindle detects stretch and trace the stretch reflex including reciprocal inhibition of the antagonist.

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3

COMP

Golgi tendon organ

You should be able to: Contrast the stimulus and the reflex response of the Golgi tendon organ with those of the muscle spindle.

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COMP

Withdrawal and crossed extensor reflexes

You should be able to: Trace the flexor withdrawal reflex and the crossed extensor reflex and explain how the two together preserve balance.

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5

COMP

Ascending and descending pathways

You should be able to: Compare the dorsal column, spinothalamic, and corticospinal pathways by the information carried, the side of decussation, and the deficit produced by a lesion.

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6

COMP

Cerebrospinal fluid and the blood brain barrier

You should be able to: State the functions of cerebrospinal fluid and explain how the blood brain barrier selects what reaches neurons, including why lipid soluble drugs cross readily.

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7

COMP

Reflex testing and reaction time

You should be able to: Elicit and grade deep tendon reflexes and measure reaction time to distinguish a reflex response from a voluntary response by latency.

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8

COMP

Sensory transduction

You should be able to: Explain how a stimulus is converted into a receptor potential and how that graded potential becomes a train of action potentials in the sensory neuron.

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9

COMP

Receptor classification

You should be able to: Classify a receptor by stimulus type as a mechanoreceptor, chemoreceptor, thermoreceptor, photoreceptor, or nociceptor and give a body location for each.

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10

COMP

Stimulus coding

You should be able to: Explain how the nervous system encodes modality, location, intensity, and duration and identify the labeled line principle.

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11

COMP

Receptive fields and acuity

You should be able to: Relate receptive field size and lateral inhibition to two point discrimination and predict which body regions have the finest acuity.

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12

COMP

Receptor adaptation

You should be able to: Distinguish tonic from phasic receptors by their adaptation rate and match each to a stimulus the body needs to monitor continuously or only at onset.

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13

COMP

Somatosensory pathways

You should be able to: Trace touch and pain information from receptor to cortex and explain why the somatosensory homunculus is disproportionate.

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COMP

Pain and its modulation

You should be able to: Distinguish fast and slow pain by fiber type, explain referred pain by convergence, and describe how gate control and endogenous opioids reduce pain transmission.

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COMP

Tactile mapping and adaptation testing

You should be able to: Measure two point discrimination at several body sites and show receptor adaptation and referred sensation experimentally.

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16

COMP

Optics and accommodation

You should be able to: Explain refraction by the cornea and lens and describe how accommodation and the pupillary reflex focus near and far images.

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COMP

Phototransduction

You should be able to: Trace phototransduction from photon absorption by retinal through the change in cyclic GMP to the decrease in glutamate release and explain why photoreceptors hyperpolarize to light.

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18

COMP

Rods cones and visual processing

You should be able to: Compare rods and cones by sensitivity and acuity and trace the visual pathway to predict the field deficit produced by a lesion at the optic nerve or the optic chiasm.

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COMP

Refractive errors and clinical vision

You should be able to: Explain myopia, hyperopia, presbyopia, and astigmatism in terms of focal point and eyeball or lens shape and state the corrective lens for each.

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COMP

Auditory transduction

You should be able to: Trace sound from the tympanic membrane through the ossicles and cochlear fluid to hair cell bending and auditory nerve firing and explain how pitch and loudness are coded.

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21

COMP

Conductive and sensorineural hearing loss

You should be able to: Distinguish conductive from sensorineural hearing loss by the site of the lesion and by the expected Weber and Rinne results.

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22

COMP

Equilibrium

You should be able to: Explain how the semicircular canals detect angular acceleration and how the utricle and saccule detect linear acceleration and head position.

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COMP

Taste and smell

You should be able to: Compare gustatory and olfactory transduction and explain why olfactory input reaches the limbic system without a thalamic relay.

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24

COMP

Vision testing

You should be able to: Perform visual acuity, blind spot, accommodation, and color vision testing and interpret each result.

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25

Hearing and equilibrium testing

COMP

You should be able to: Perform Weber and Rinne tuning fork tests and a balance test and interpret the findings.

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26

COMP

Motor control hierarchy

You should be able to: Order the levels of motor control from spinal reflex through brainstem to cortex and state what each level contributes to a voluntary movement.

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COMP

Corticospinal pathway

You should be able to: Trace the corticospinal pathway from the primary motor cortex to the skeletal muscle and identify where it crosses.

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COMP

Cerebellum and basal ganglia

You should be able to: Contrast the contributions of the cerebellum and the basal ganglia to movement and match a described motor sign to the structure involved.

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COMP

Upper and lower motor neuron signs

You should be able to: Distinguish upper from lower motor neuron lesions by tone, reflexes, and muscle bulk and assign a set of findings to the correct level.

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COMP

Autonomic and somatic organization

You should be able to: Contrast the somatic motor system with the autonomic two neuron pathway by neuron count and by effector tissue.

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COMP

Sympathetic and parasympathetic divisions

You should be able to: Compare the two autonomic divisions by CNS origin and ganglion location and predict the effect of each on heart rate and on the gastrointestinal tract.

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COMP

Autonomic neurotransmitters and receptors

You should be able to: Match nicotinic, muscarinic, alpha, and beta receptors to their neurotransmitter, their location, and the response each produces.

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COMP

Dual innervation and tonic control

You should be able to: Explain dual innervation and autonomic tone and predict the effect of removing one division from a dually innervated organ.

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COMP

Autonomic pharmacology

You should be able to: Predict the effect of a beta blocker or a muscarinic antagonist on a named organ from the receptor it targets.

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COMP

Autonomic function testing

You should be able to: Measure heart rate variability and the response to a cold pressor or orthostatic challenge and interpret the result as sympathetic or parasympathetic dominance.

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